COMPUTATIONAL MEDICINE

Brain Tumour Surgical Planning for High Grade Glioma

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INTRODUCTION: | Dynamics of Surgical Planning

High Grade Glioma Diagnosis

- Irregular tumour progression patterns
- Tumour cells can infiltrate brain tissue beyond observable with conventional MRI
- Neurosurgical team must establish tumour extent and determine amount of tissue removable without causing significant neurological impairment.

INTRODUCTION: | **MRI Suitability and Pitfalls**

- Visualize details of tumour anatomy, tissue characteristics, vascularity without invasive procedure.
- MRI + AI/ML techniques supports delineation of tumour regions, computes volume and quantitative feature extraction that support surgical planning.
- Only suggests presence of glioma but not definitive establishment of histological and molecular diagnosis.
- Insufficient for examination of Gliobastoma, IDH-wildtype diffuse gliomas.

INTRODUCTION: | **Adopted Workflow**

1. Biological sampling and diagnostic investigations.
2. Conventional MRI imaging and examination
3. Medica Image processing (Tumour segmentation and Quantitative feature extraction)

Segmentation & Feature extraction Data Source:
Medical Segmentation Decathlon, “Task01_BrainTumour” from Hugging Face derived from BraTs (Brain Tumour Segmentation) research data

STEP ONE: | Biological Samples

Brain Tissue Biopsy

Stereotactic Needle Biopsy:

- Small tissue sample obtained using needle guided by pre-operative CT or MRI.
- Used when tumour is located in deep brain structures e.g. thalamus, where extensive surgical removal is more risky.

Limitation:

High-grade gliomas are heterogenous; small sample tissue gives limited information.

STEP ONE: | Biological Samples

Brain Tissue Biopsy

Surgical Resection Specimen:

- Obtain as much tissue as can be safely achieved, for examination.
- More sample for comprehensive diagnosis.

Limitation:

Obtaining large portions of brain tissue potentially impairs neurological function.

STEP ONE: | # Histopathological Examination

Process & Pathological Feature Identification

Process:

- Tissue embedded in Paraffin, stained with Haematoxylin & Eosin, then examined under Microscopy.

Feature Identification:

- Increased cellularity.
- Nuclear atypia & Cellular pleomorphism
- Increased mitotic activity
- Microvascular proliferation
- Areas of tumour necrosis

World Health Organization (WHO) Recommendation:
Diagnosis of CNS tumours must place considerable emphasis on molecular & histological characteristics.

STEP TWO: | MRI Imaging Features for Treatment Outcome Predictions

- MRI guides in examining Tumour extent, Tissue infiltration and mapping removal tissue during surgery.
- Conventional and Advanced MRI is employed.

Conventional MRI features:

- **T1-weighted contrast enhanced imaging:** helps estimate visible tumour.
- **T2-weighted imaging:** Provides info about tissue surrounding the tumour.
- **Fluid-Attenuated Inversion Recovery(FLAIR):** Visualization of peritumoral hyperintensity
- **Diffusion-Weighted Imaging(DWI) & Apparent Diffusion Coefficient (ADC):**
  - DWI: Measures movement of water molecules within tissue
  - ADC: Provides info about water diffusion.
  - Regions with dense tumour restrict diffusion.

STEP TWO: | MRI Imaging Features for Treatment Outcome Predictions

Advanced MRI techniques:

- Magnetic Resonance Spectroscopy: Examines chemical composition of tissues
- Diffusion Tensor Imaging and Functional MRI: Determine safest surgical method and extent of resection without unacceptable neurological damage.

STEP TWO: | MRI Imaging Features for Treatment Outcome Predictions

MRI features Associated with treatment outcomes:

Feature	Relation with Outcome	Relevance to planning
Total tumour volume	Large tumour ≈ Poor prognosis & Treatment complexity	Estimates overall extent of disease.
Enhancing Tumour Volume	Reflects volume of contrast enhancing abnormality	Quantify visible tumour burden
Necrotic fraction	Proportion of tumour core with necrotic/dead tissue.	Characterizes tumour heterogeneity.
FLAIR Abnormality Volume	Reveals oedema and Infiltrative tumour beyond enhanced region	Reveals wider extent of abnormal tissue.

STEP TWO: | MRI Imaging Features for Treatment Outcome Predictions

MRI features Associated with treatment outcomes:

Feature	Relation with Outcome	Relevance to planning
ADC Values	Low ADC ≈ Increased cellularity	Shows regions with different tissue characteristics.
Relative Cerebral Blood Volume (rCBV)	Increased values ≈ Angiogenesis and Vascularity	Shows biologically active tumour tissue
Tumour Location	Tumours in critical regions restrict surgical resection	Assessment of surgical accessibility and functional risk.
Residual Tumour Volume	Greater post-operative residual tumour ≈ poorer outcomes	Assesses effectiveness of resection.

STEP THREE: |

Image Segmentation and Feature Extraction

Image Data Source:
BraTs: Medical Segmentation Decathlon (MSD), Task01: Brain Tumours

Programming Languages Employed:
Python, R

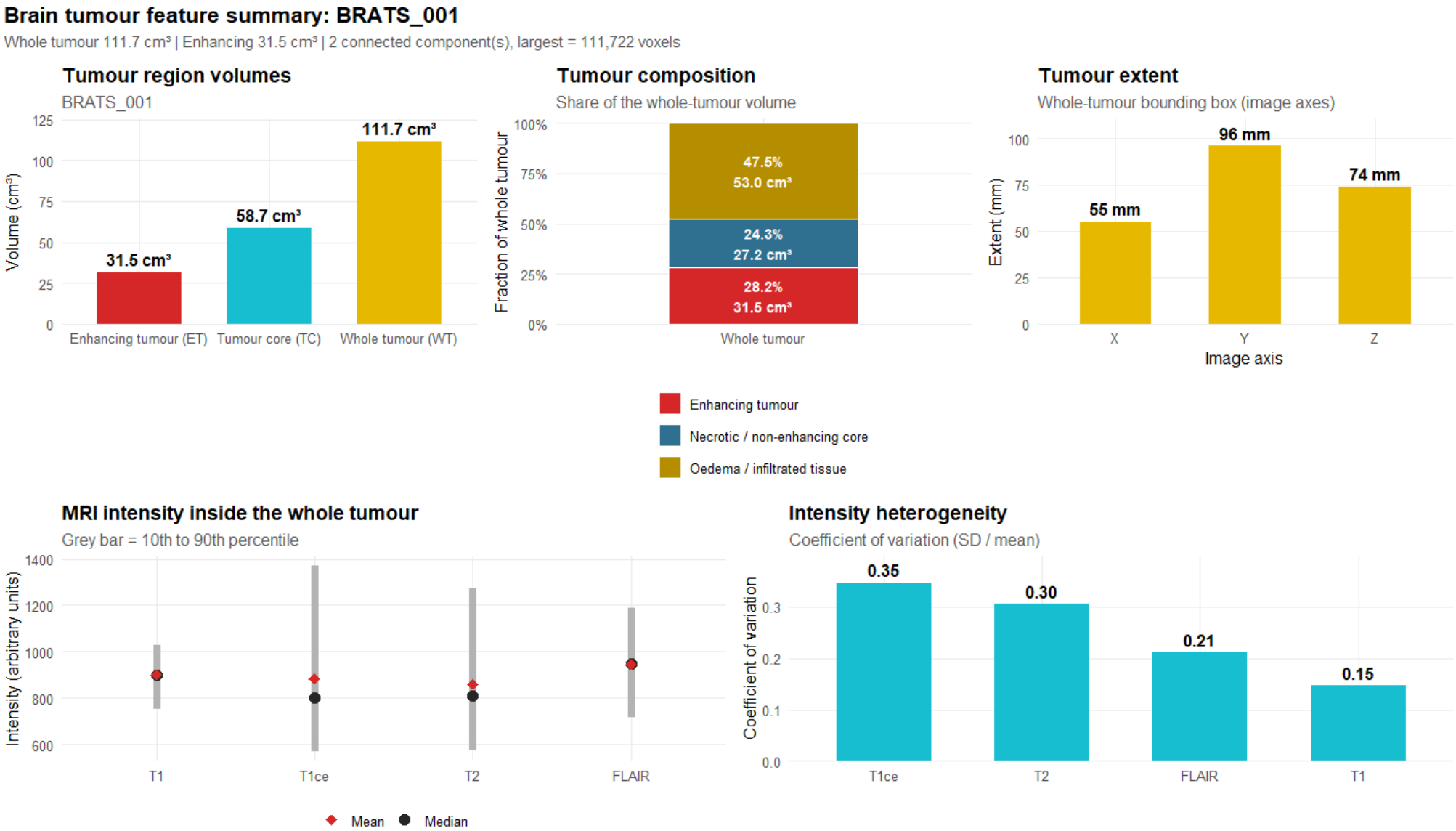
Packages:
Python (nibabel, numpy, pandas, matplotlib, scipy, scikit-learn, scikit-image)
R(readr, ggplot2, dplyr, tidyr)

STEP THREE: | Image Segmentation and Feature Extraction

Considering image BraTs_001.nii.gz: Extracted Sample Features

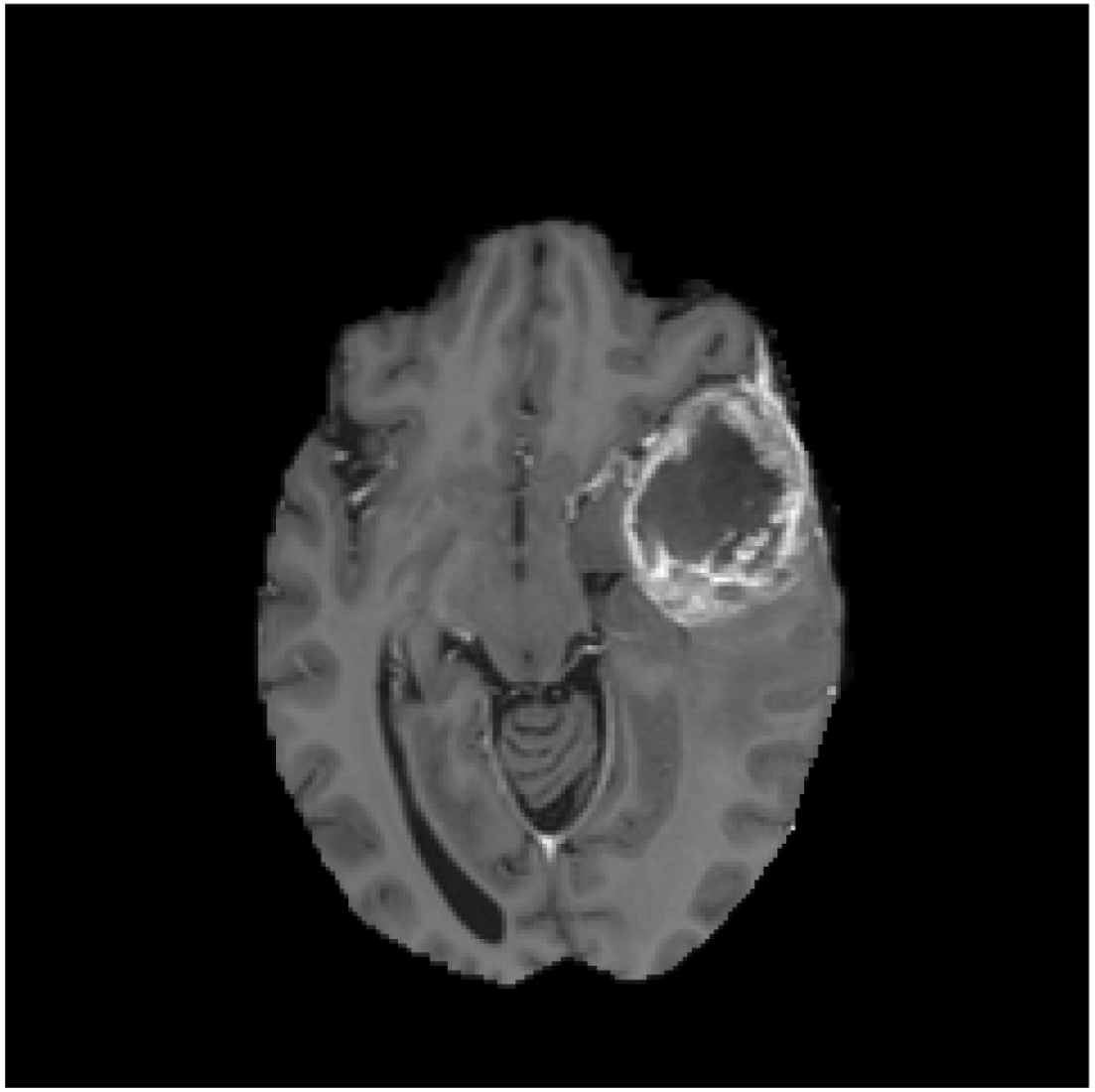
Measure/Feature	Value
Whole Tumour (WT) volume	111.724 cm^3
Tumour Core (TC) volume	58.674 cm^3
Enhancing Tumour volume	31.485 cm^3
Non-Enhancing/Necrotic Core volume	= (TC - ET) = 27.189 cm^3
Oedema/Infiltrated Tissue Volume	= $[{}^{47}/_{100} \times WT]$ = 52.510 cm^3
Components	2

STEP THREE: | Feature Extraction

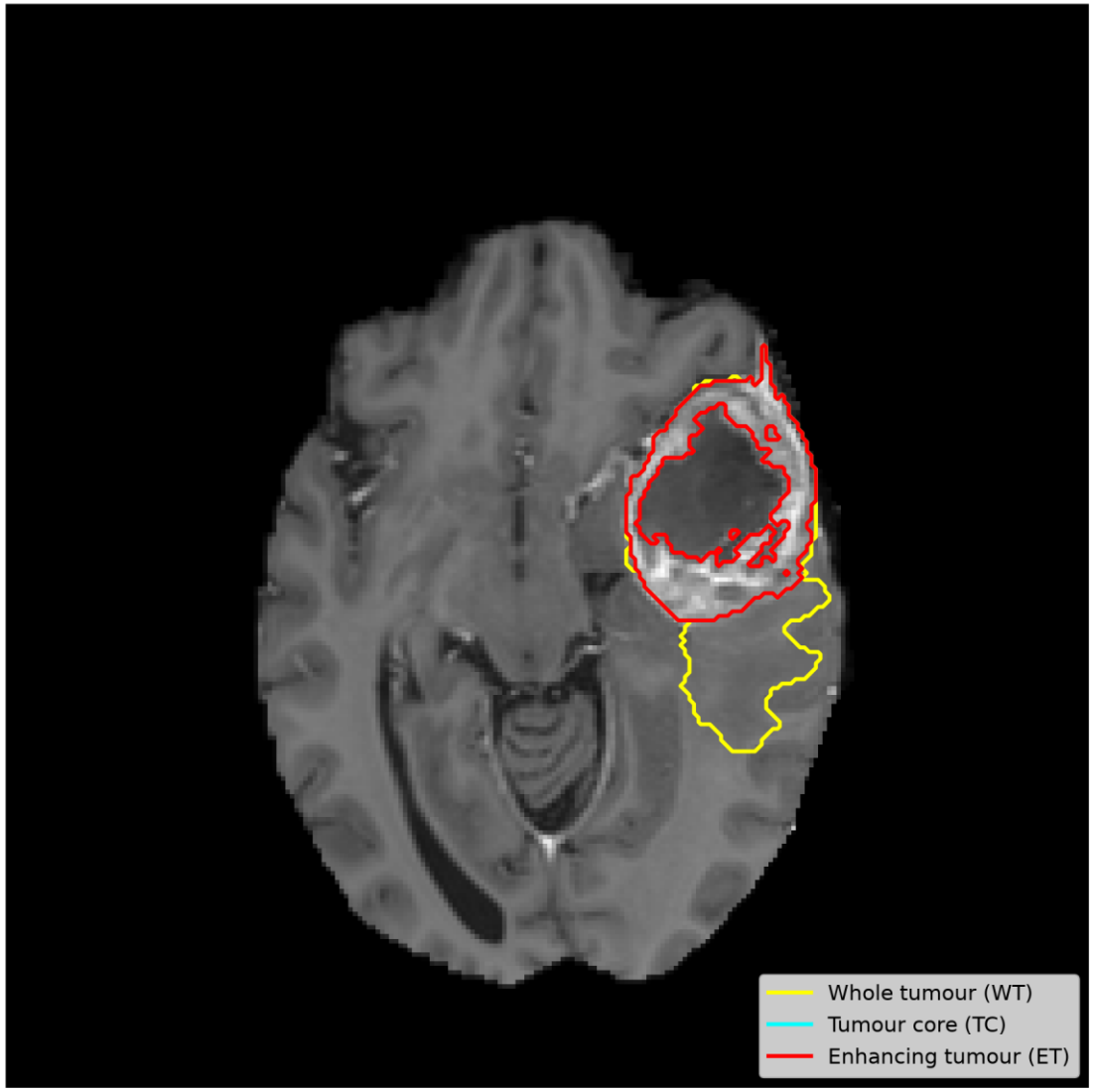


STEP THREE: | **Feature Segmentation**

BRATS_001: T1ce, axial slice 63



Tumour segmentation: WT, TC and ET



STEP THREE: |

Image Segmentation and Feature Extraction

Intuition:

- T1ce (T1-Weighted Contrast Enhanced Region) surrounding dark necrotic center indicates suspected high-grade glioma, but not conclusively.
- Other conditions e.g. Abscess, Metastasis and Lymphoma can produce similar results when imaged.
- Histological and Molecular tests are necessary for a conclusive diagnosis.

STEP THREE: |

Image Segmentation and Feature Extraction

Contribution of Analysis to Surgical Planning:

- Enhancing volume is baseline for extent of surgical resection. It has prognostic importance.
- Large necrotic core is non-diagnostic. Active rim must be targeted.
- Size of tumour raises perioperative risks which need closer attention the to patient
- Location of tumour guides neurosurgical team on pre-operative checks to minimize unacceptable functional impairment to patient.

WE CLOSE HERE: | **Thank You!**